

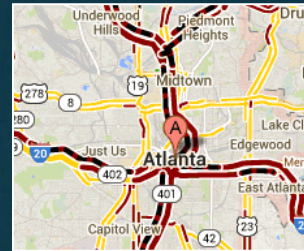
DATA MINING INTRODUCTION

Large-scale Data is Everywhere!

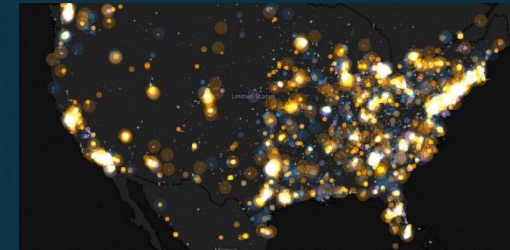
- There has been enormous data growth in both commercial and scientific databases due to advances in data generation and collection technologies
- New mantra
 - Gather whatever data you can whenever and wherever possible.
- Expectations
 - Gathered data will have value either for the purpose collected or for a purpose not envisioned.



E-Commerce



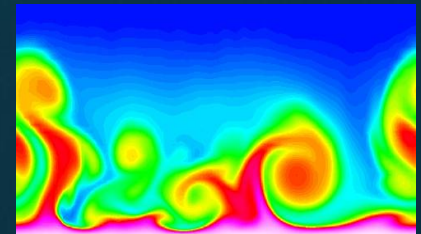
Traffic Patterns



Social Networking: Twitter



Sensor Networks



Computational Simulations

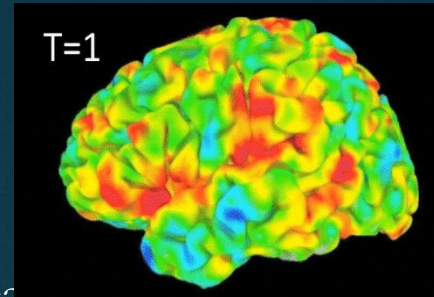
Why Data Mining? Commercial Viewpoint

- Lots of data is being collected and warehoused
 - Web data
 - Yahoo has Peta Bytes of web data
 - Facebook has billions of active users
 - purchases at department/grocery stores, e-commerce
 - Amazon handles millions of visits/day
 - Bank/Credit Card transactions
- Computers have become cheaper and more powerful
- Competitive Pressure is Strong
 - Provide better, customized services for an edge (e.g. in Customer Relationship Management)



Why Data Mining? Scientific Viewpoint

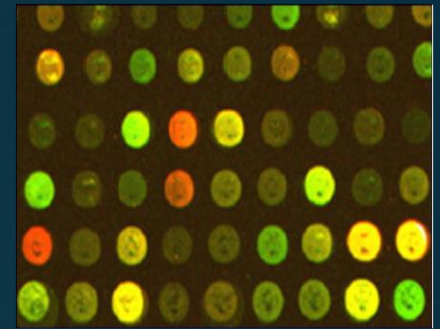
- Data collected and stored at enormous speeds
 - remote sensors on a satellite
 - NASA EOSDIS archives over petabytes of earth science data / year
 - telescopes scanning the skies
 - Sky survey data
 - High-throughput biological data
 - Scientific simulations
 - terabytes of data generated in a few hours
- Data mining helps scientists
 - in automated analysis of massive datasets
 - in hypothesis formation



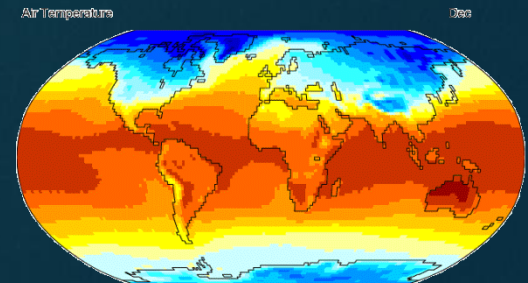
fMRI Data from Brain



Sky Survey Data



Gene Expression Data

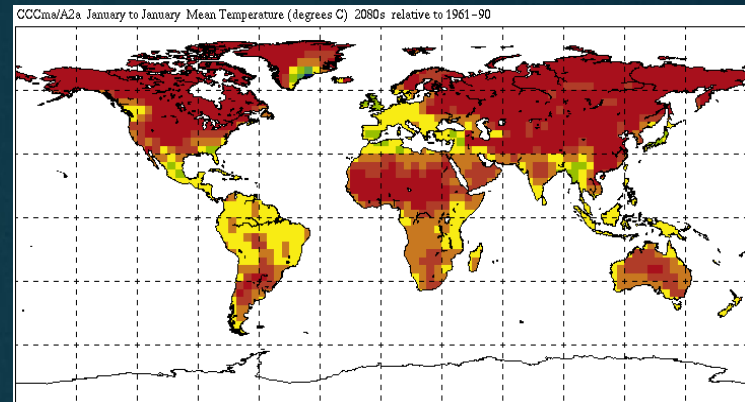


Surface Temperature of Earth

Great Opportunities to Solve Society's Major Problems



Improving health care and reducing costs



Predicting the impact of climate change



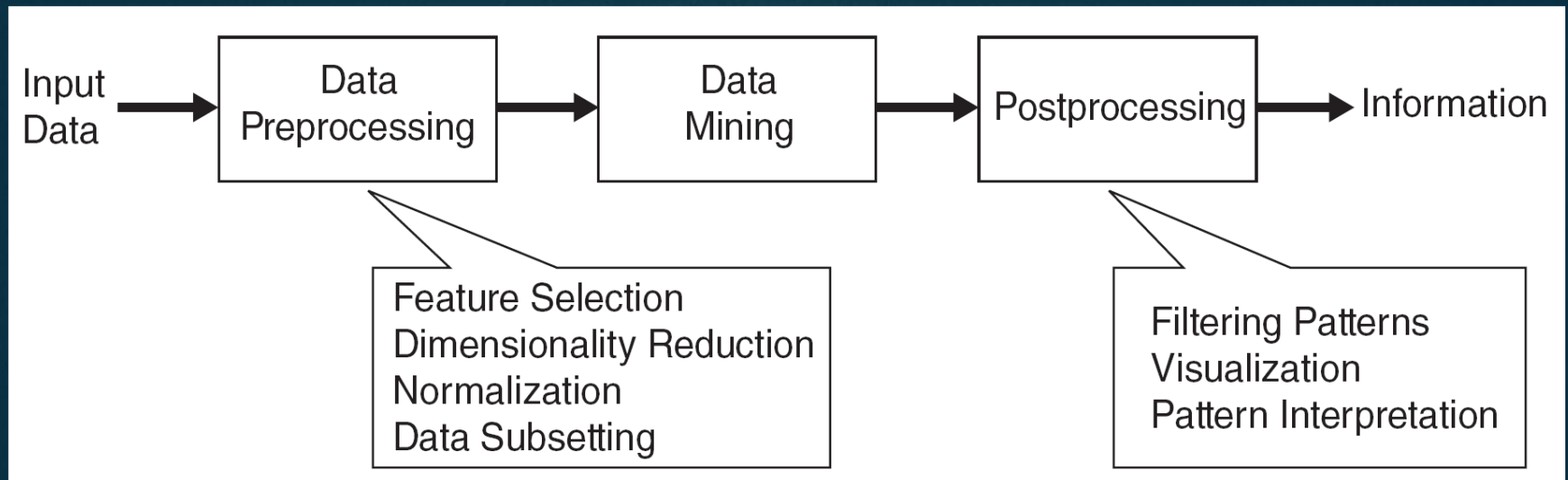
Finding alternative/ green energy sources



Reducing hunger and poverty by increasing agriculture production

What is Data Mining?

- Non-trivial extraction of implicit, previously unknown and potentially useful information from data
- Exploration & analysis, by automatic or semi-automatic means, of large quantities of data in order to discover meaningful patterns



What is (not) Data Mining?

□ What is not Data Mining?

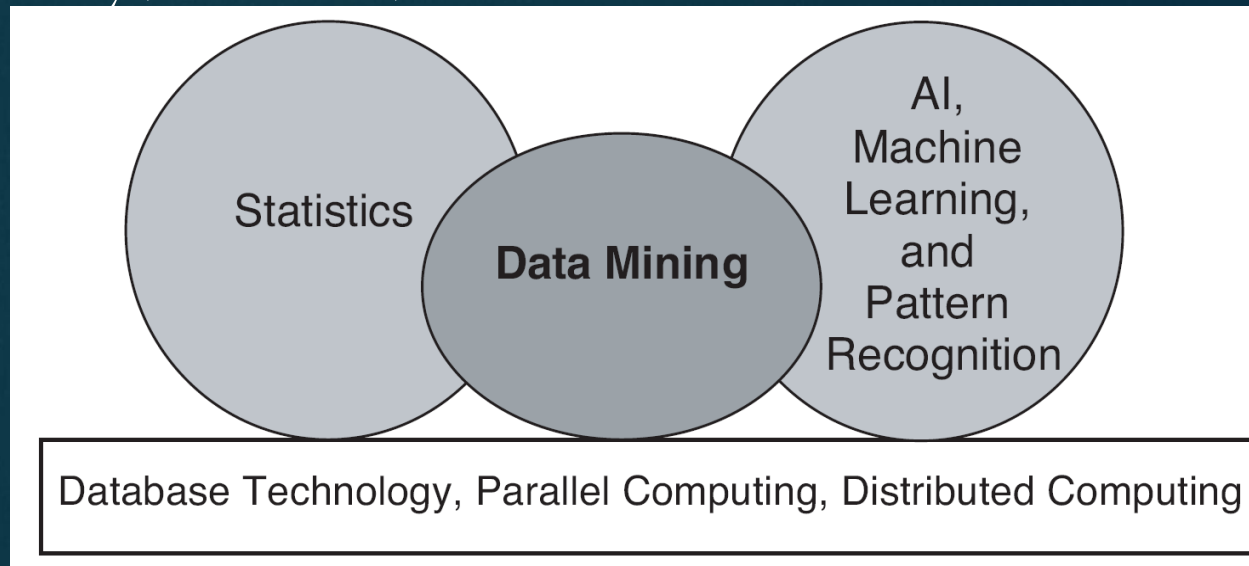
- Look up phone number in phone directory
- Query a Web search engine for information about “Amazon”

□ What is Data Mining?

- Certain names are more prevalent in certain US locations (O’Brien, O’Rourke, O’Reilly... in Boston area)
- Group together similar documents returned by search engine according to their context (e.g., Amazon rainforest, Amazon.com)

Origins of Data Mining

- Draws ideas from machine learning/AI, pattern recognition, statistics, and database systems
- Traditional techniques may be unsuitable due to data that is
 - Large-scale
 - High dimensional
 - Heterogeneous
 - Complex
 - Distributed
- A key component of the emerging field of data science and data-driven discovery



Motivating Challenges

- Scalability
- High Dimensionality
- Heterogeneous and Complex Data
- Data Ownership and Distribution
- Non-traditional Analysis

Scalability

- Data sets with sizes of gigabytes, terabytes, or even petabytes are becoming common
- Data mining algorithms are scalable
- DM algorithms employ special search strategies to handle exponential search problems
- Implements novel data structures to access individual records in an efficient manner
- Can be improved by using sampling or developing parallel and distributed algorithms

High Dimensionality

- Data sets with large number of attributes are common
- E.g: In bioinformatics, progress in microarray technology has produced gene expression data involving thousands of features
- Data sets with temporal or spatial components also tend to have high dimensionality

Heterogeneous and Complex Data

- Traditional data analysis methods often deal with data sets containing attributes of the same type, either continuous or categorical
- But need for techniques that can handle heterogeneous attributes has grown
- Emergence of more complex data objects
- Examples non-traditional types of data include
 - collections of Web pages containing semi-structured text and hyperlinks;
 - DNA data with sequential and three-dimensional structure
 - climate data

Data ownership and Distribution

- Data is geographically distributed among resources belonging to multiple entities
- Distributed data mining algorithms challenges:
 - how to reduce the amount of communication needed to perform the distributed computation
 - how to effectively consolidate the data mining results obtained from multiple source
 - how to address data security issues

Non-traditional Analysis

- Hypothesis is proposed, an experiment is designed to gather the data, and then the data is analyzed with respect to the hypothesis- labor intensive
- Current data analysis tasks often require the generation and evaluation of thousands of hypotheses
- Development of some data mining techniques has been motivated by the desire to automate the process of hypothesis generation and evaluation
- Data sets analyzed in data mining are typically not the result of a carefully designed experiment
- Often represent opportunistic samples of the data
- Data sets frequently involve non-traditional types of data and data distributions

Data Mining Tasks

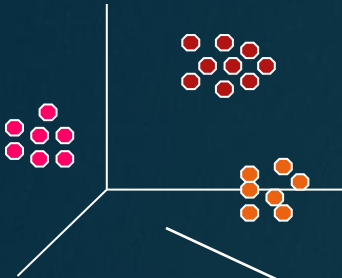
► Predictive Methods

- The objective of these tasks is to predict the value of a particular attribute based on the values of other attributes
- Use some variables to predict unknown or future values of other variables
- The attribute to be predicted is commonly known as the target or dependent variable
- The attributes used for making the prediction are known as the explanatory or independent variables

► Descriptive Methods

- objective is to derive patterns (correlations, trends, clusters, trajectories, and anomalies) that summarize the underlying relationships in data
- Often exploratory in nature and frequently require postprocessing techniques to validate and explain the results
- Find human-interpretable patterns that describe the data

Data Mining Tasks

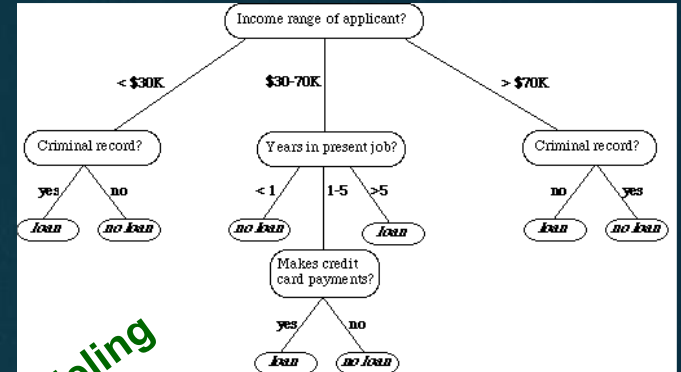


Clustering

Data

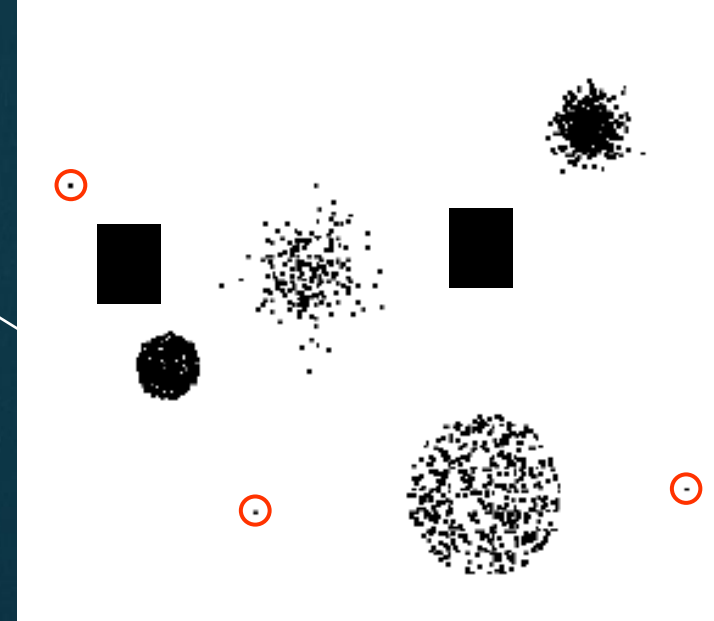
Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes
11	No	Married	60K	No
12	Yes	Divorced	220K	No
13	No	Single	85K	Yes
14	No	Married	75K	No
15	No	Single	90K	Yes

Association Rules



Predictive Modeling

Anomaly Detection



Predictive Modeling

- ▶ Task of building a model for the target variable as a function of the explanatory variables
- ▶ **Classification**
 - Used for discrete target variables
 - Predicting whether a Web user will make a purchase at an online bookstore is a classification task because the target variable is binary-valued
- ▶ **Regression**
 - used for continuous target variables
 - forecasting the future price of a stock is a regression task because price is a continuous-valued attribute.

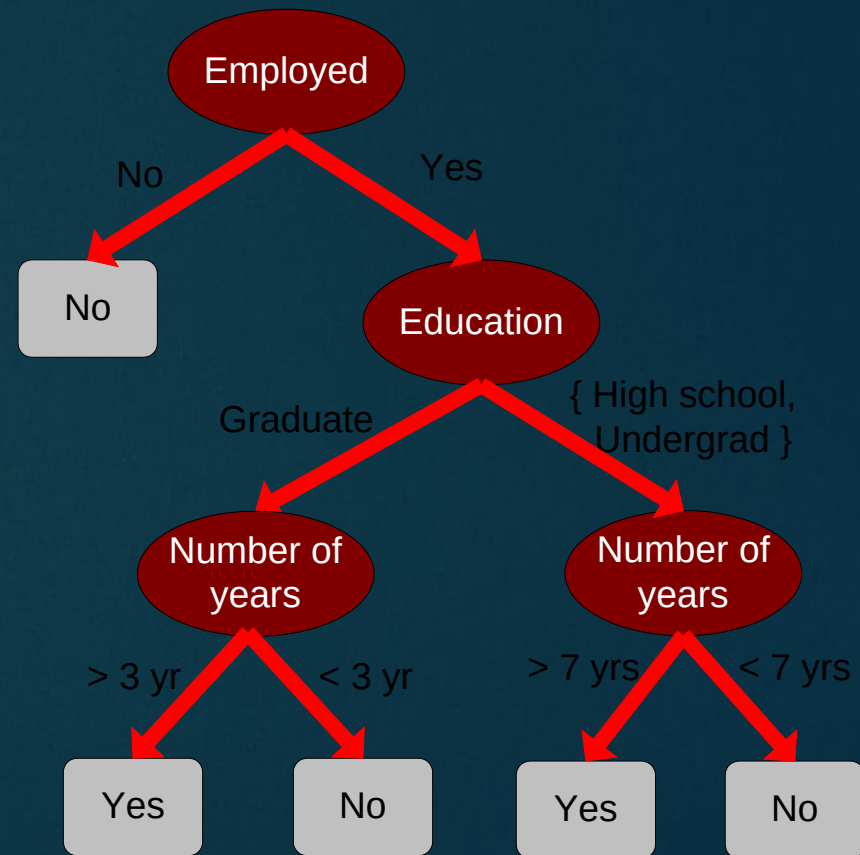
Predictive Modeling: Classification

- Find a model for class attribute as a function of the values of other attributes

Class

<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
...	...	...	...	...

Model for predicting credit worthiness

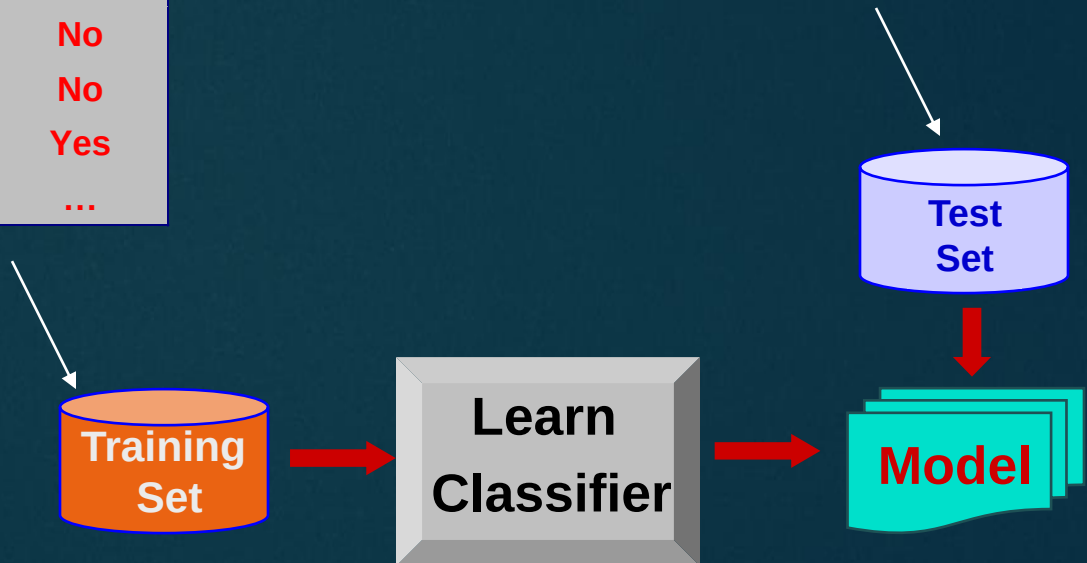


Classification Example

categorical categorical quantitative class

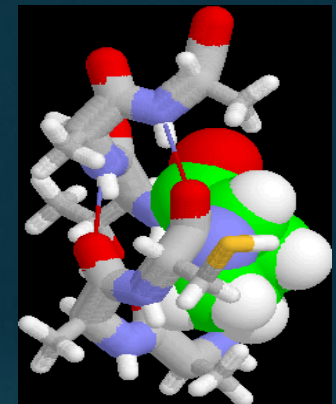
<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
...	...	...	...	...

<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Undergrad	7	?
2	No	Graduate	3	?
3	Yes	High School	2	?
...	...	...	...	...



Examples of Classification Task

- Classifying credit card transactions as legitimate or fraudulent
- Classifying land covers (water bodies, urban areas, forests, etc.) using satellite data
- Categorizing news stories as finance, weather, entertainment, sports, etc
- Identifying intruders in the cyberspace
- Predicting tumor cells as benign or malignant
- Classifying secondary structures of protein as alpha-helix, beta-sheet, or random coil

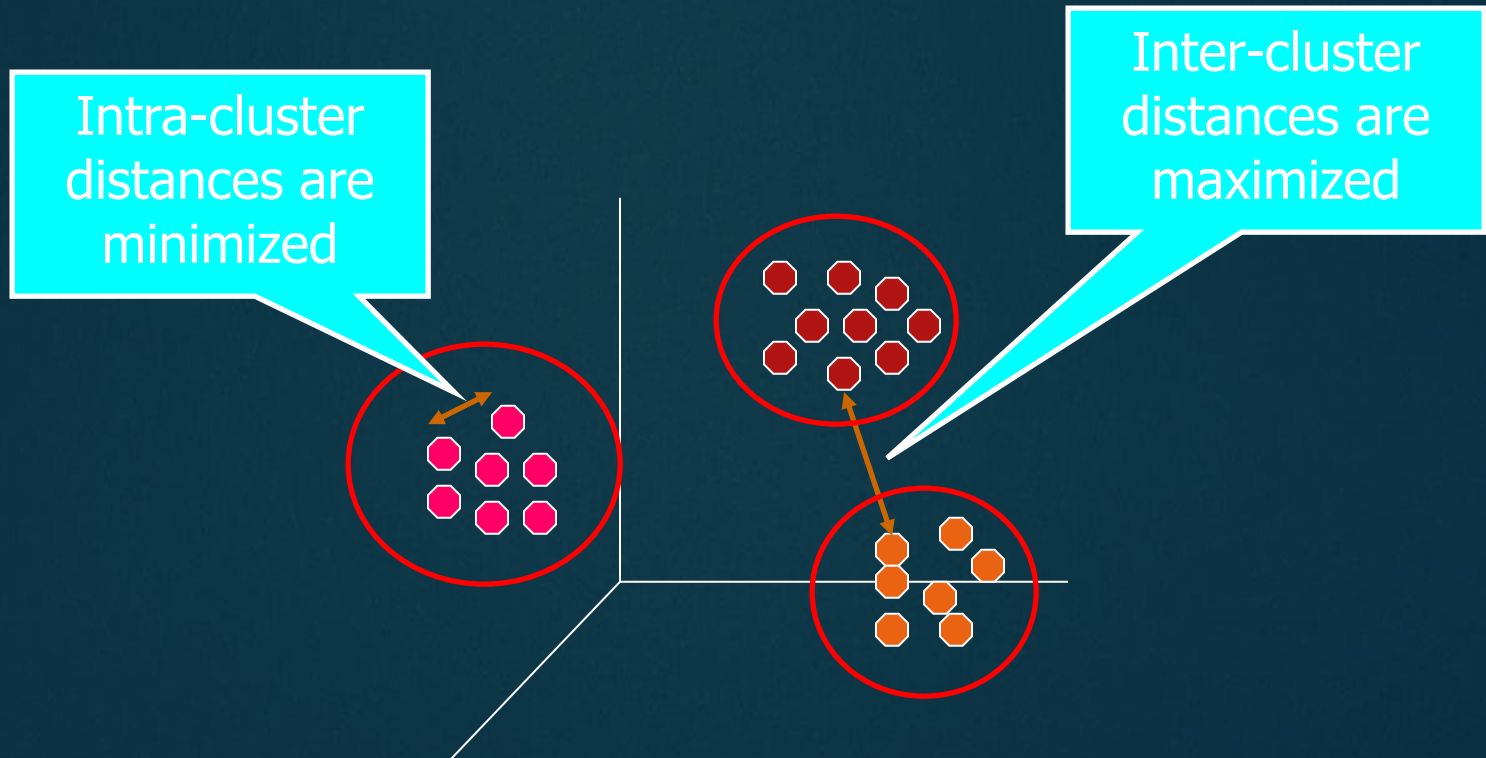


Regression

- Predict a value of a given continuous valued variable based on the values of other variables, assuming a linear or nonlinear model of dependency.
- Extensively studied in statistics, neural network fields.
- Examples:
 - Predicting sales amounts of new product based on advertising expenditure.
 - Predicting wind velocities as a function of temperature, humidity, air pressure, etc.
 - Time series prediction of stock market indices.

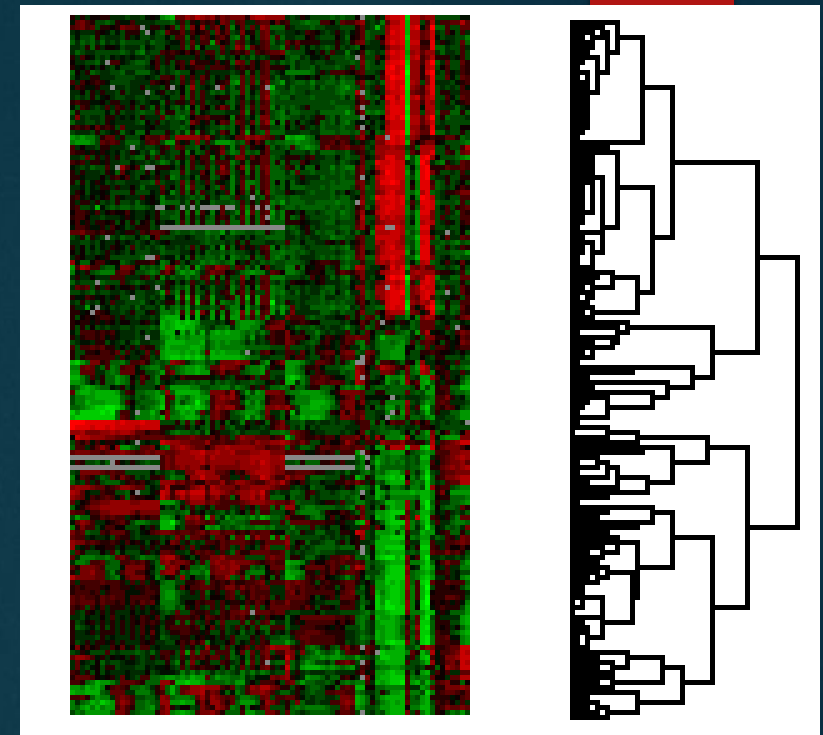
Clustering

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups

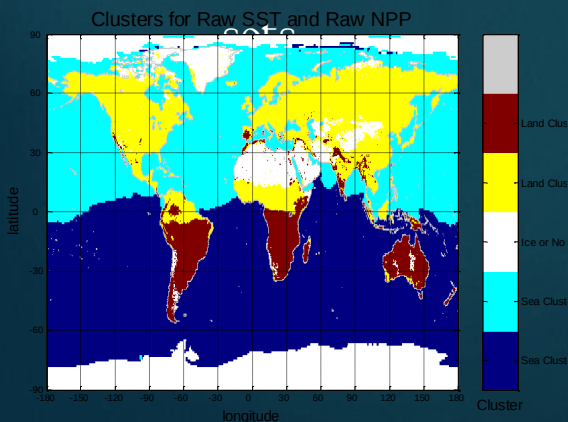


Applications of Cluster Analysis

- Understanding
 - Custom profiling for targeted marketing
 - Group related documents for browsing
 - Group genes and proteins that have similar functionality
 - Group stocks with similar price fluctuations
- Summarization
 - Reduce the size of large data



Courtesy: Michael Eisen



Use of K-means to partition Sea Surface Temperature (SST) and Net Primary Production (NPP) into clusters that reflect the Northern and Southern Hemispheres.



Association analysis

- Used to discover patterns that describe strongly associated features in the data.
- The discovered patterns are typically represented in the form of implication rules or feature subsets
- Because of the exponential size of its search space, the goal of association analysis is to extract the most interesting patterns in an efficient manner

Association Rule Discovery: Definition

- Given a set of records each of which contain some number of items from a given collection
 - Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

$\{\text{Milk}\} \rightarrow \{\text{Coke}\}$

$\{\text{Diaper, Milk}\} \rightarrow \{\text{Beer}\}$

Association Analysis: Applications

- Market-basket analysis
 - Rules are used for sales promotion, shelf management, and inventory management
- Telecommunication alarm diagnosis
 - Rules are used to find combination of alarms that occur together frequently in the same time period
- Medical Informatics
 - Rules are used to find combination of patient symptoms and test results associated with certain diseases

Deviation/Anomaly/Change Detection

- Detect significant deviations from normal behavior
- Applications:
 - Credit Card Fraud Detection
 - Network Intrusion Detection
 - Identify anomalous behavior from sensor networks for monitoring and surveillance.
 - Detecting changes in the global forest cover.

